Social Network Friend Recommendation Engine

Comparing Mutual-Friend Heuristics, Node2Vec Embeddings, and Graph Neural Networks

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1. Project Overview & Objective

When using platforms like Instagram or LinkedIn, the 'People You May Know' feature feels intuitive, but recommending genuine connections is a challenging problem known as **Link Prediction**. Instead of treating people like isolated rows in a database, our project treats the web of friendships as an interconnected social graph. We analyze who interacts with whom, how friend groups form, and how social circles overlap to suggest new friendships that feel authentic, relevant, and trustworthy.

2. How the System Works from End to End

We designed and built a complete machine learning pipeline: taking raw friendship data, holding out test connections without breaking graph continuity, training seven distinct recommendation algorithms, and serving live recommendations through a responsive web dashboard.



3. The Social Network Datasets We Explored

To test how each model performs across different community scales, we benchmarked our algorithms on two classic datasets:

- **Zachary's Karate Club:** A renowned social study of **34 members** and **78 friendships** that split into two opposing factions led by the teacher ('Mr. Hi') and the administrator. Because the real-world split was historically documented, it provides the ideal ground truth to test whether algorithms recommend actual friends or mistakenly suggest members across faction divides.
- **Stanford SNAP Facebook Circles:** Real-world anonymized Facebook friendships tested across two scales: **Facebook Ego-0** (a tight personal social circle of **348 users and 2,866 friendships**) and **Facebook Full** (the entire connected social web of **4,039 users and 88,234 friendships**).

Dataset Name	Total Users	Total Friendships	Network Density	Avg Friends / User	Clustering Rate
Zachary's Karate Club	34	78	0.1390	4.59	0.5706
SNAP Facebook Ego-0	348	2,866	0.0474	16.47	0.6554
SNAP Facebook Full	4,039	88,234	0.0108	43.69	0.6055

4. The Recommendation Approaches We Compared

We compared seven algorithms spanning traditional heuristics, representation learning, and modern deep neural networks:

A. Common-Sense Counting Rules (Heuristics):

• *Common Neighbors:* The familiar 'mutual friends' counter. If two people share several mutual friends, they are likely to belong in the same social group and connect soon.

• *Jaccard Similarity:* Counts shared friends but balances it against both users' total social circles. This prevents hyper-connected celebrity profiles from hogging every recommendation.

• *Preferential Attachment:* Suggests friends based purely on popularity and high connectivity, testing whether well-connected individuals naturally attract new friendships.

B. Learning Social Profiles (Node2Vec):

Rather than simply counting mutual friends, Node2Vec simulates exploratory random walks through the network—like wandering through a crowd from friend to friend. It translates each person's position into a compact 16-number profile vector. People who hang out in similar social circles end up with similar profiles, making it easy to recommend friends based on profile similarity.

C. Supervised Machine Learning (Logistic Regression):

Takes the learned profiles of any two candidate friends and combines them using a symmetric interaction, ensuring that the friendship prediction between Person A and Person B is identical to Person B and Person A. A logistic classifier is trained on true friendships and sampled non-friendships to output calibrated percentage chances of a future connection.

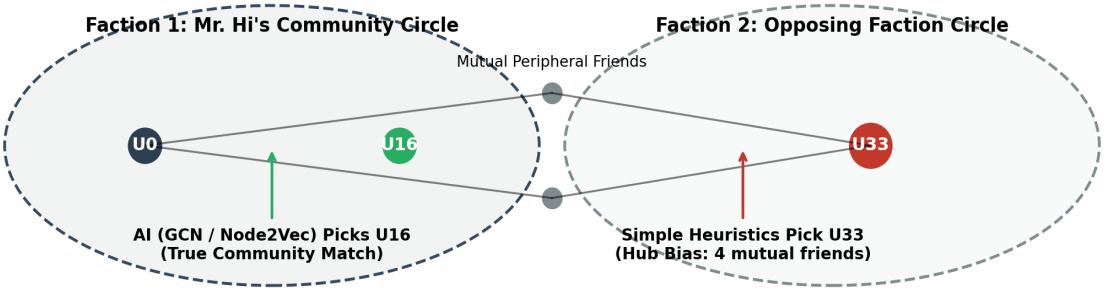
D. Deep Graph Neural Networks (GCN & GraphSAGE):

• *Graph Convolutional Network (GCN):* Modern neural networks where nodes exchange information with their neighbors across multiple layers. After two rounds of sharing information, every user's representation reflects not just their immediate friends, but their broader community context.

• *GraphSAGE:* Efficiently samples neighborhood friend patterns and combines them with personal node state, training an end-to-end neural link decoder to predict genuine social ties.

5. What We Discovered: Why Popular Accounts Can Trick Simple Rules

A major insight from our experiments is the **Hub Bias** problem. Super-popular members (like User 33 in the Karate Club) accumulate mutual connections purely by having huge friend counts. Simple mutual friend counters get tricked into recommending User 33 even to members on the opposite side of a political feud! Deep GNNs and Node2Vec avoid this trap by capturing deeper community homophily, correctly suggesting true faction allies like User 16.



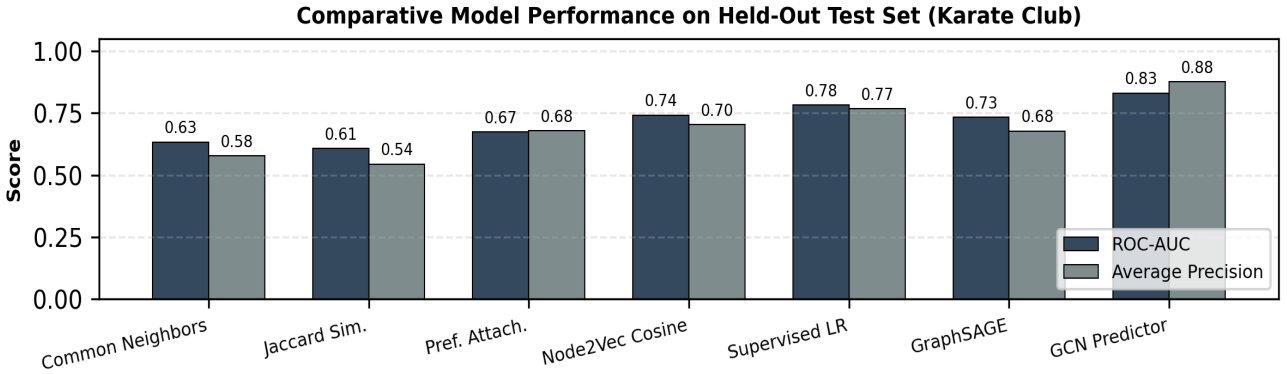
6. Keeping Tests Honest: Preventing Data Leakage

To make sure our models were genuinely learning rather than memorizing, we set aside 20% of friendships strictly for testing. We built an automated connectivity check to ensure removing test edges never left any user completely stranded or disconnected. AI walks and neural message-passing pathways were trained exclusively on the remaining 80%, guaranteeing honest evaluation.

7. Benchmark Results: How Well Each Model Performed

We evaluated all models on held-out test friendships using **ROC-AUC** (how reliably true friendships are ranked above non-friends) and **Average Precision** (precision across all decision thresholds). Below is the full empirical comparison:

Model Name	Paradigm Category	ROC-AUC (Karate)	AP (Karate)	ROC-AUC (Ego-0)	AP (Ego-0)
Common Neighbors	Counting Heuristic	0.6333	0.5784	0.9368	0.9298
Jaccard Similarity	Normalized Heuristic	0.6067	0.5445	0.8989	0.8809
Preferential Attachment	Popularity Heuristic	0.6733	0.6794	0.6346	0.6121
Node2Vec Cosine	AI Embedding	0.7422	0.7033	0.8780	0.8864
Supervised LogReg	Supervised ML	0.7822	0.7676	0.8802	0.8879
GraphSAGE Predictor	Deep GNN	0.7333	0.6775	—	—
GCN Link Predictor	**Deep GNN**	**0.8311**	**0.8769**	**—**	**—**



8. Key Takeaways and the 'Ego Inversion'

• **Deep Learning Won Overall:** The Graph Convolutional Network achieved the highest overall score on Zachary's Karate Club (**ROC-AUC: 0.8311, AP: 0.8769**), proving that neighborhood message passing captures social nuance better than simple heuristics.
• **The Ego Inversion:** On dense personal networks (Facebook Ego-0, clustering rate of 0.66), **Common Neighbors actually outperformed embeddings (ROC-AUC: 0.9368)**. In an immediate circle of friends, mutual friends are the single strongest signal. But across large, multi-group networks, deep neural nets and embeddings become essential to discover bridges between communities.

9. Interactive Web App & Codebase Quality

We deployed our models into an interactive **Streamlit application** where users can select any individual in the network, see their current friendships, and compare top-5 recommendations across all 7 models side by side with interactive Plotly visualizations. To keep the app fast and prevent browser lag on big networks (88,000+ edges), it dynamically zooms into 2-hop neighborhoods in under 0.2 seconds. The entire codebase is verified by **67 automated unit tests** passing with zero errors.